

KOUSUMI MONDAL

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PROFESSIONAL SUMMARY

Final-year Computer Science Engineer specializing in **AI-powered solutions** and **Machine Learning**. Practical experience includes developing a **Retrieval-Augmented Generation platform** and an image denoising model, achieving high-quality image restoration through deep learning techniques. Proficient in **Python**, **OpenCV**, and **Natural Language Processing**, ready to contribute as a Python AI/ML & Generative AI Intern.

WORK EXPERIENCE

Hackathon: Honeywell Campus Connect Hackathon Jul 2026 - Jul 2026

Honeywell Remote

- Designed a hybrid **AI-powered SOC anomaly detection system** using a Transformer Autoencoder and LightGBM, integrating **10-event behavioral sequences** and **10+ engineered security features**.
- Developed an interactive **Streamlit SOC dashboard** with **4 analyst modules**, AI risk scoring, SHAP explainability, and ranked alert prioritization to improve the effectiveness of security analysts.
- Achieved **99.44% accuracy**, **91.4% recall**, **89.67% AUPRC**, and **82.73% F1-score** for impossible-travel attack detection through an end-to-end anomaly detection and evaluation pipeline, utilizing **deterministic logic** and ensuring **cost-efficient AI pipelines**.

Technologies / Skills Used : Python, Pytorch, Transformers, deterministic logic, cost-efficient AI pipelines

PROJECTS

MedInsight – Medical Intelligence System | GitHub Website Mar 2026 - May 2026

- Architected a production-grade **Retrieval-Augmented Generation (RAG)** platform using **LangChain**, **Groq-hosted LLMs**, and **ChromaDB** to deliver context-aware clinical question answering over biomedical literature. **Applied decision analytics** and **optimization techniques** to enhance performance.
- Designed an end-to-end **semantic retrieval layer** with a chunking strategy, embedding pipeline, and vector-index tuning to boost retrieval precision and minimize response latency across extensive research corpora.
- Developed agentic workflows and deployed a FastAPI backend and React frontend, leveraging Biopython for structured extraction and parsing of biological research data, demonstrating comprehensive **MLOps practices** from data ingestion to inference.
- Authored and maintained **product documentation** for both customers and internal teams.

Technologies / Tools Used : Python, LangChain, RAG, ChromaDB, decision analytics, optimization techniques, agentic workflows, graph-based retrieval, tool calling, state management, custom evaluation frameworks

Image Denoiser using DRUNet | GitHub Website Jul 2026 - Present

- Developed a **DRUNet-based** image denoising model on the DIV2K dataset, achieving **high-quality image restoration** through deep residual learning techniques and **OpenCV**.
- Built an end-to-end training pipeline featuring realistic noise injection, data augmentation, and optimization strategies to enhance **model generalization**.
- Evaluated model performance using PSNR and SSIM metrics while optimizing inference workflows for **efficient processing** of high-resolution images, incorporating **Natural Language Processing** techniques as needed.

Technologies / Tools Used : Pytorch, Transformers, Docker, Next.js, OpenCV, Natural Language Processing

LexTrace – Legal Document Intelligence System | Website Mar 2026 - Apr 2026

- Engineered a **full-stack RAG pipeline for legal document intelligence**, unifying ingestion of scanned and digital documents via **PyMuPDF** and **Tesseract OCR**.
- Implemented semantic retrieval using **Sentence Transformers** and **MongoDB vector search**, reducing query latency while improving retrieval accuracy across large legal corpora.
- Integrated a **Groq-hosted LLM** through **LangChain** to deliver citation-aware legal question answering, prioritizing traceability and factual grounding — a core requirement for enterprise legal-tech systems.
- Built a responsive TypeScript frontend, delivering a production-ready, end-to-end legal-tech application from architecture to UI.

Technologies / Tools Used : FastAPI, LangChain, Groq, MongoDB

SKILLS

Databases :	PostgreSQL, MySQL, MongoDB, Supabase
Frameworks & Libraries :	PyTorch, Scikit-learn, Transformers, LangChain, LangGraph
Languages :	English, Hindi, Bengali
Programming Languages :	Python, Java, SQL, JavaScript
CS Fundamentals :	Data Structures & Algorithms, Databases, System Design Fundamentals
Tools & Platforms :	Power BI, Tableau, Excel, MongoDB, AWS, Docker, CI/CD, FastAPI, Figma, Claude, Lovable, Git/Github

EDUCATIONS

Vellore Institute of Technology 2023 - 2027

B.Tech Computer Science & Engineering CGPA : 8.11

- Relevant Coursework: **Artificial Intelligence**, Machine Learning, Marketing Analytics, Distributed Systems, Data Structures and Algorithms.

AWARDS & ACHIEVEMENTS

- Selected from **800 finalist teams** out of **31,000+ registered participants** nationwide in the Meta X PyTorch OpenEnv Hackathon.